Report B — Automation Report

What we automated, what it can do, what is new, and how to grow it.

Automation means: a student or mentor does not have to click Load → Preprocess → Reconstruct for every file. One command (or one GUI button) finds files, picks scans, runs the same scientific pipeline, scores the result with written thresholds, and writes a report. The science is the same as the GUI. The novelty is the unattended path around it.

Not a clinical diagnosis. Pass/fail is a research quality gate, not a tumor diagnosis.

1. What problem automation solves

UM-BMID-style cubes can contain hundreds of scans. Doing each one by hand is slow and uneven (different people pick different beamformers). Automation makes the experiment repeatable: same geometry grid, same ROI rules, same score weights, same report layout.

- Discover measurement vs metadata files under one or more folders.
- Pair md_list_*.mat with fd_data_*.mat so people do not upload labels without data.
- Choose a diverse subset of BMID scans (healthy + small/medium/large + location/BIRADS/shape).
- Run load → preprocess → DAS/DMAS/DMAS-D4 → ROI spots → tumor-candidate → characterization → confidence.
- Classify each case pass / warning / fail / blocked using a named profile.
- Write latest_auto_analysis.md, JSON evidence, and showcase PNG figures.
- Optionally open the best tumor-candidate case in the GUI.

2. How you can run it (three doors, one engine)

All three call AutoValidationService. That is important: CLI, GUI, and MCP cannot drift apart.

- **CLI:** python tools/run_auto_validation.py --profile balanced
- **GUI:** Acquisition → Run Auto Validation (scope: 8 diverse scans by default).
- **MCP:** automation/mcp_server.py exposes the same job over stdio for agent tools.
- Related helper: python tools/auto_intake_agent.py checks files before a person uploads them.

Default roots prefer microwave/umbmid/simple-clean when that folder exists. Otherwise project/datasets is used (that is why a recent run validated sample_touchstone.s2p).

3. Capability map — module by module

3.1 Discovery (automation/discovery.py)

Walks folders. Classifies each file as measurement, metadata, or unknown. fd_data_*.mat is a UM-BMID cube. md_list_* / metadata_*.mat is labels only. .s2p/.s1p/.s4p/.s8p and other .mat files are measurements. Unknown extensions are skipped. Ranking prefers fd_data_ files first so real cubes beat leftover samples when both exist.

3.2 Resolver (automation/resolver.py)

Turns files into targets: (measurement path, optional scan index, optional metadata path). Metadata-only files are useful only if a paired fd_data file can be found beside them or in the repo. BMID strategies:

- **diverse** (default) — about 8 scans: one healthy, one small, one medium, one large, then extra quadrants / BIRADS / shapes.
- **all** — every scan up to max_bmid_scans (0 means unlimited; this is slow for ~2000 scans).

- **tumor_and_healthy** — strongest labeled tumor plus one healthy.
- **first / first_healthy** — simple debugging picks.

Tumor ranking when not using diverse: larger diameter first, then an offset preference near ~3 cm so huge edge phantoms are not always chosen.

3.3 Per-target pipeline (automation/pipeline.py)

This is the automated twin of the GUI workflow.

- Load with load_dataset(path, scan_index).
- Read ground-truth tumor (x,y) from metadata if present.
- Preprocess: BMID → all methods none, Week 3 off. Otherwise → Savitzky–Golay window 7 poly 3, Week 3 on, clutter on, global normalize on.
- Reconstruct DAS, DMAS, DMAS-D4 on a 64×64 grid. BMID FOV forced to ±6 cm; phase-delay radius off for comparable batch runs.
- Select beamformer: if GT exists, pick the image whose ROI is closest to the labeled tumor; else weighted quality.
- Detect up to 4 ROIs (threshold 0.7 internally, min_score_ratio 0.28 in automation). If GT exists, keep the nearest hotspot even if NMS dropped it.
- High-resolution ROI reconstruct (min 96^2, up to 256^2, ×4 upscale of the coarse box).
- Characterize size/SCR/FWHM; assess confidence; write figures.
- Classify with the chosen threshold profile.

3.4 Quality profiles (automation/policy.py + classifier.py)

Hard fails: load error, non-finite image, energy below min, ROI area below min. Soft warnings: localization error, SCR, SNR, confidence, or “GT tumor but not a candidate”. Blocked: cannot even start (no paired measurement).

Gate	strict	balanced	lenient
min energy	1e-10	1e-12	1e-14
min ROI area (px)	8	4	1
max loc. error (cm)	1.5	3.0	5.0
min SCR	5.0	2.0	0.5
min SNR	1.0	0.5	0.1
min confidence	0.55	0.35	0.20
require candidate if GT	yes	yes	no
finite output required	yes	yes	yes

Warning cases can still be recommended_upload if localization is within 1.25× the error gate. That lets slightly blurry but useful scans into a demo set.

3.5 Reports, cache, figures, MCP

- reports.py writes results/latest_auto_analysis.md (tumor index, detection table, showcase, walkthrough) and latest_auto_validation.json.
- Showcase pick: labeled tumor first, then strongest candidate, then smaller localization error, then higher confidence.
- figures.py copies selected / beamformer-compare / ROI-refine PNGs.
- cache.py avoids re-running identical (file, scan, profile) work in one session.
- MCP server is a thin stdio wrapper so an external agent can start the same job.

4. What automation is capable of today (plain list)

- Batch-screen a folder of .mat / .s2p files without opening each GUI tab.
- Tell you which BMID indices are labeled tumors before reconstruction (tumor index).
- Produce Yes/No tumor-candidate plus a reason sentence per case.

- Report localization error in cm against laboratory GT when labels exist.
- Compare DAS vs DMAS vs DMAS-D4 automatically.
- Keep a JSON audit trail of metrics, taxonomy, preprocessing steps, and ROI spots.
- Fail closed on corrupted files instead of crashing the desktop app.
- Run load-only mode (--no-reconstruction) for a fast finite/energy sanity check.

What it does **not** do: it is not a trained neural classifier; taxonomy is metadata buckets. It does not claim BIRADS medically. It does not replace a radiologist. It does not search every geometry on every scan (that is Auto Tweak in the GUI; batch uses a fixed conservative geometry).

5. Novelty this automation brings to the project

Many microwave-imaging student projects stop at “we plotted one DAS image.” The new layer is an experiment operating system around the physics:

- **Format-aware batching:** the same job understands Touchstone S-matrices and BMID cubes, including “do not preprocess _adi”.
- **Label-aware scan picking:** diverse strategy is not random; it forces healthy + size + location coverage so a demo is honest.
- **Physics-in-the-loop scoring:** gates are localization (cm), SCR, SNR, ROI area, energy — not a black-box accuracy percentage.
- **Multi-spot ROI:** more than one hotspot can be reported; GT is used to keep a nearby spot, not to hide other spots.
- **One evidence pack:** markdown for humans, JSON for machines, PNG for slides, GUI handoff for inspection.
- **Profiled strictness:** mentors can ask for lenient triage or strict localization without changing code.
- **Three interfaces, one pipeline:** students, GUI users, and MCP agents cannot silently use different math.

6. How this sits next to the rest of the software

Manual GUI path remains the teaching path: you see every tab. Automation is the verification path: you prove the same modules work on many files. Auto Tweak (reconstruction/auto_calibrate.py) is a different automation: it searches antenna angle, flip, 355° vs 360°, phase-delay radius, and ROI mode for **one** loaded scan. Batch validation does not currently call Auto Tweak, on purpose, so a 8-scan run finishes in a reasonable time and stays comparable.

7. How we can develop it further

Near term (practical)

- Default Module 3 input to Hamming / Week-3 time-domain when those arrays exist (README already lists this gap).
- Optional “Auto Tweak lite” inside batch: only on showcase tumor cases, not on every healthy scan.
- Write a CSV summary (one row per scan) for Excel mentors who will not open JSON.
- Store per-scan reconstruction settings in the JSON so a GUI reload is bit-identical.
- Add a dry-run that only prints the tumor index and chosen diverse scans (no beamforming).

Medium term (science)

- Replace the placeholder severity mapping with a trained model that uses image features + metadata, still labeled research-only.
- Learn suspicion_threshold per dataset family instead of a global 0.45.
- Support empty-chamber (_emp) subtraction when both adi and emp cubes are present.
- Quantify false-alarm rate on healthy scans as a first-class metric in the analysis report.
- Add a third data family (e.g. time-domain .mat already IFFT’d) with an explicit skip-IFFT flag.

Longer term (product / teaching)

- A simple “lab exam mode”: hidden GT, student reconstructs, automation grades localization only.
- Cloud/batch runner that processes the full 2264-scan inventory overnight and publishes a dashboard.
- Uncertainty maps (pixel-wise coherence factor) exported beside the image.
- Keep the medical disclaimer in every new report surface so nobody treats scores as diagnosis.

8. Beginner cheat-sheet

- If you have one .s2p: automation will load it, clean S21, reconstruct, and likely say tumor-candidate Unknown (no GT).
- If you have UM-BMID: put fd_data and md_list together; run diverse profile balanced.
- Read section 1 of latest_auto_analysis.md to find tumor scan indices.
- Read section 2 for Yes/No and localization error.
- Open the showcase image instead of clicking every tab.
- Use lenient only when you are debugging failures; use strict before a demo you care about.

Tumour classification status

Official first-scan score: **71/100**. This is the nested test on UM-BMID (signal plus image, extra training scans from phantoms that are not in the test fold, cut-off chosen on training phantoms only). It has not been replaced.

Official follow-up score: **97/100**. The scan is compared with earlier healthy scans of the same breast from the same sitting. Tumour and healthy cases are given the same number of those scans, at most 3, so the count cannot give the answer away.

A later side check scored about **69/100**. That check used only the signal features and only the fair training scans. Adding scattering-spectrum and frequency-slope formulas did not raise it. It is recorded as a side check, not as a new official score.

A convolutional network on the breast-window sinogram, and a PCA-then-logistic classifier on that same sinogram, were then scored on the fair phantom-held-out test. They scored **50/100** and **45/100**. The network could fit the breasts it trained on and did not carry that fit to a new phantom. Neither replaced the official first-scan model.

Two write-ups live in results/reports/classification/: **how_classification_works.pdf** (process, cut-offs and formulas, in plain language) and **guide_progress.pdf** (numbers and progress for a project guide). Machine-readable scores stay in results/classification/final_summary.json.